CRICKSTATS

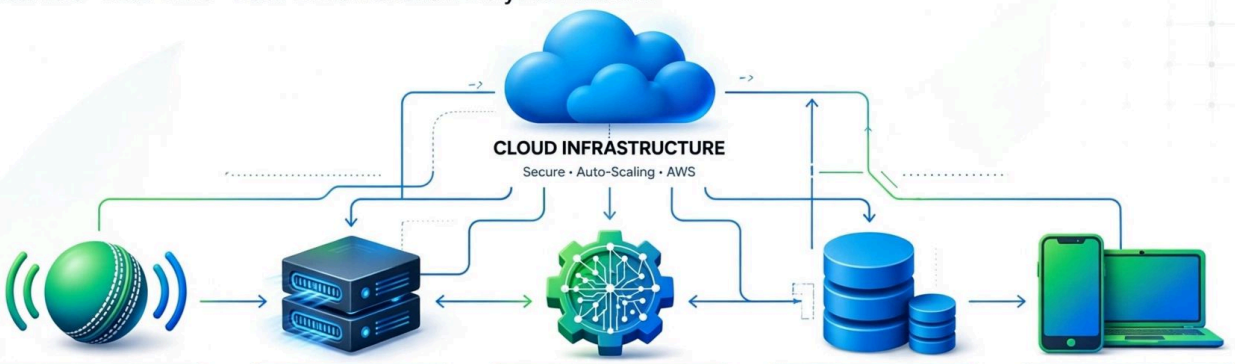
MODULE 3 • PROJECT DESIGN

Slide 1/5

MODULE 3 - PROJECT DESIGN

Step 1: Overall Solution Design

Scalable • Real-Time • Data-Driven Cricket Analytics Platform



DATA COLLECTION
Live Match Data • Player Stats • Sensors

API GATEWAY
REST API • Data Ingestion • Secure Auth

PROCESSING ENGINE
ML Analytics • Real-Time Processing • Insights

STATS DATABASE
PostgreSQL • Match History • Performance Data

USER INTERFACES
Web Dashboard • Mobile App

Real-Time Analytics | Low Latency Processing | Secure & Reliable

KEY FOCUS: MODULAR • SCALABLE • SPORTS TECH

1 / 5

CRICKSTATS

MODULE 3 • PROJECT DESIGN

MODULE 3 - PROJECT DESIGN

Step 2: Define Objects

Designing Core Data Objects for Player & Match Performance

Player Object

Core Entity • Profile Data

- **Name**
  String • Full player name
- **Age**
  Integer • Player age in years
- **Team**
  String • Current team
- **Role**
  Enum • Batsman • Bowler • All-Rounder • Keeper
- **Jersey**
  Integer • Jersey number
- **Photo**
  URL • Player profile photo

Key: Unique PlayerID • Indexed by Team

Match Performance Object

Performance • Match Stats

- **Runs**
  Integer • Runs scored in match
- **Balls**
  Integer • Balls faced/bowled
- **Wickets**
  Integer • Wickets taken
- **Overs**
  Decimal • Overs bowled
- **Catches**
  Integer • Catches taken

Relation: Linked to PlayerID • Linked to MatchID

Relationship: 1 Player → Many Match Performances

Stored in Stats Database

Validated • Type-Safe • Scalable

KEY FOCUS: MODULAR • SCALABLE • SPORTS TECH

CRICKSTATS

MODULE 3 • PROJECT DESIGN

Slide 3/5

MODULE 3 - PROJECT DESIGN

Step 3: Object Relationships

Mapping Connections Between Player & Match Performance Objects



Player Object

Primary Entity

PK: PlayerID
Unique Identifier

- **Name**
- Age
- Team • Role • Jersey • Photo
- Photo

Cardinality: ONE (1)

ENTITY RELATIONSHIP DIAGRAM

ONE-TO-MANY RELATIONSHIP

A single Player can have many Match Performances

1 → ∞



Match Performance Object

Child Entity

PK: PerformanceID
Unique Identifier

- **FK: PlayerID**
- Linked to Player
- Runs • Balls • Wickets • Overs • Catches
- MatchDate

Cardinality: MANY (*)

LOOKUP RELATIONSHIP

Match Performance → Player via PlayerID
Foreign Key

RELATION BEHAVIOR

Cascade Delete • Referential Integrity Enabled

QUERY EXAMPLE

Get All Performances
WHERE PlayerID = P123

DESIGN BENEFIT: Enables historical performance tracking per player → Supports analytics, stats aggregation, and player profiles

KEY FOCUS: MODULAR • SCALABLE • SPORTS TECH

3 / 5

CRICKSTATS

MODULE 3 • PROJECT DESIGN

Slide 4/5

MODULE 3 - PROJECT DESIGN

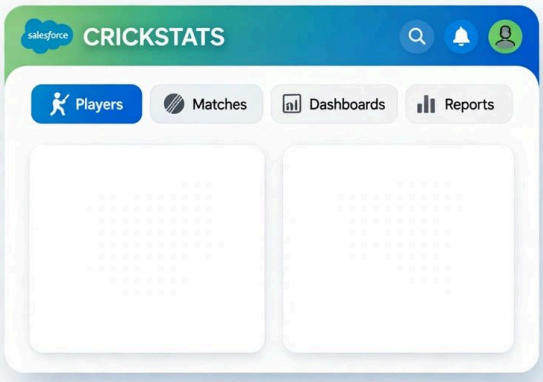
Step 4: Lightning App Architecture

Building the Salesforce Lightning Application for CRICKSTATS

Powered by Salesforce Lightning

LIGHTNING COMPONENTS

- **Player List Component •** Dynamic list view of players
- **Performance Chart Component** Runs & Wickets Visualization
- **Match Filter Component •** Filter by Date, Team, Role



Players Tab • Overview

Live Data

Player	Role	Team	Avg Runs	Actions
Virat K	Batsman	India	45.2	View

Total Players: 120 | Active Matches: 18 | Avg Strike Rate: 132.5

LIGHTNING UI LAYER
Lightning Web Components → Responsive UI → Mobile Ready

APEX & AUTOMATION
Apex Controllers • Flows • Validation Rules

DATA LAYER
Custom Objects → SOQL External API Integration

KEY ADVANTAGES: No-Code → Low-Code → Real-Time Dashboarding → Mobile-First Experience → Secure & Role-Based Access

KEY FOCUS: MODULAR • SCALABLE • SPORTS TECH

4 / 5

CRICKSTATS

MODULE 3 • PROJECT DESIGN

Slide 5/5

MODULE 3 - PROJECT DESIGN

Step 5: Dashboards, Reports & Flows

Analytics, Insights & Automation for CRICKSTATS Success

DASHBOARDS & REPORTS

Total Runs Scored
12,450
↑ +12.5% vs Last Month

Wickets Taken
387
↑ +8.2% vs Last Month

Top Batters — Leaderboard

1. Virat K — 850 Runs
2. Rohit S — 720 Runs
3. Gill P — 610 Runs
4. SKY — 540 Runs
5. Rahul K — 480 Runs

Runs Over Time — Line Chart



Aug • Sep • Oct • Nov • Dec

Total Runs

AUTOMATED FLOWS

WHEN:
New Match Performance Created

FLOW CONDITION:
Runs > 50?

YES → Action:
Update Player Stats

Trigger:
Send Congrats Notification to Player

NO → Action:
Log for Review

Update Report
Dashboard Log

Daily Summary Flow • Runs Every 9PM

Automated Email Report - Weekly Performance Report Sent to Coach

OUTCOMES: Data-Driven Insights • Real-Time Reporting • Automated Updates – Enhanced Decision Making

KEY FOCUS: MODULAR • SCALABLE • SPORTS TECH

FINAL - MODULE 3 COMPLETE

5 / 5